

Foundations of Regularization machine learning

[Regularization machine learning]

1.0 Content Block

$$\min_f \sum_{i=1}^n V(f(\hat{x}_i), y_i) + \lambda \|f\|_{\mathcal{H}}^2 \quad \{\displaystyle \min_{f \in \mathcal{H}} \sum_{i=1}^n V(f(\hat{x}_i), y_i) + \lambda \|f\|_{\mathcal{H}}^2\}$$

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$$\min_{f \in \mathcal{R}} \|f\|_{\mathcal{L}} = \min_{f \in \mathcal{R}} \left\{ \|f\|_{\mathcal{L}}^2 = \sum_{i=1}^n \|f(x_i) - y_i\|^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\} \quad \{\displaystyle \min_{f \in \mathcal{R}} \|f\|_{\mathcal{L}} = \min_{f \in \mathcal{R}} \left\{ \|f\|_{\mathcal{L}}^2 = \sum_{i=1}^n \|f(x_i) - y_i\|^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\}\}$$

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$$R(f_1, \dots, f_T) = \sum_{t=1}^T \|f_t - f_s\|_{\mathcal{M}}^2 \quad \{\displaystyle R(f_1, \dots, f_T) = \sum_{t=1}^T \|f_t - f_s\|_{\mathcal{M}}^2 \text{ for a given symmetric similarity matrix } M\}$$

4.0 Content Block

This can be viewed as inducing a regularizer over the L_2 norm over members of each group followed by an L_1 norm over groups. This can be solved by the proximal method, where the proximal operator is a block-wise soft-thresholding function:

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Regularization can be motivated as a technique to improve the generalizability of a learned model. The goal of this learning problem is to find a function that fits or predicts the outcome (label) that minimizes the expected error over all possible inputs and labels. The expected error of a function f_n is:

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These techniques are named for Andrey Nikolayevich Tikhonov, who applied regularization to integral equations and made important contributions in many other areas. When learning a linear function f , characterized by an unknown vector w such that $f(x) = w \cdot x$, one can add the L_2 -norm of the vector w to the loss expression in order to prefer solutions with smaller norms. Tikhonov regularization is one of the most common forms. It is also known as ridge regression. It is expressed as:

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$$R(w) = \inf_{\{w_g\}} \sum_{g=1}^G \|w_g\|_{\mathcal{L}}^2 : w = \sum_{g=1}^G w_g \quad \{\displaystyle R(w) = \inf_{\{w_g\}} \sum_{g=1}^G \|w_g\|_{\mathcal{L}}^2 : w = \sum_{g=1}^G w_g\}$$

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$$w_T = \frac{\gamma}{n} \sum_{i=0}^{T-1} (I - \gamma X^T X)^i X^T Y \quad \{\displaystyle w_T = \frac{\gamma}{n} \sum_{i=0}^{T-1} (I - \gamma X^T X)^i X^T Y\}$$

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When labels are more expensive to gather than input examples, semi-supervised learning can be useful. Regularizers have been designed to guide learning algorithms to learn models that respect the structure of

unsupervised training samples. If a symmetric weight matrix W is given, a regularizer can be defined:

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If W_{ij} encodes the result of some distance metric for points x_i and x_j , it is desirable that $f(x_i) \approx f(x_j)$. This regularizer captures this intuition, and is equivalent to:

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$S_\lambda(v) = \begin{cases} v - \lambda, & \text{if } v > \lambda \\ 0, & \text{if } v \in [-\lambda, \lambda] \\ v + \lambda, & \text{if } v < -\lambda \end{cases}$

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Enforcing a sparsity constraint on w can lead to simpler and more interpretable models. This is useful in many real-life applications such as computational biology. An example is developing a simple predictive test for a disease in order to minimize the cost of performing medical tests while maximizing predictive power. A sensible sparsity constraint is the L_0 norm $\|w\|_0$, defined as the number of non-zero elements in w . Solving a L_0 regularized learning problem, however, has been demonstrated to be NP-hard. The L_1 norm (see also Norms) can be used to approximate the optimal L_0 norm via convex relaxation. It can be shown that the L_1 norm induces sparsity. In the case of least squares, this problem is known as LASSO in statistics and basis pursuit in signal processing.

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$\text{prox}_R(v) = \arg\min_{w \in \mathbb{R}^D} \{R(w) + \frac{1}{2}\|w - v\|^2\}$, $\text{prox}_R(v) = \mathop{\text{argmin}}_{w \in \mathbb{R}^D} \{R(w) + \frac{1}{2}\|w - v\|^2\}$

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Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

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$\min_w \sum_{i=1}^n V(x_i \cdot w, y_i) + \lambda \|w\|_2^2$, $\min_w \sum_{i=1}^n V(x_i \cdot w, y_i) + \lambda \|w\|_2^2$

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$R(w) = \sum_{g=1}^G \|w_g\|_2$, where $w_g = \sum_{j=1}^{|G_g|} |w_{g_j}|$, $\|w_g\|_2 = \sqrt{\sum_{j=1}^{|G_g|} w_{g_j}^2}$

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$R(f) = \bar{f}^T L \bar{f}$ where $L = D - W$ is the Laplacian matrix of the graph induced by W . The optimization problem $\min$

$f \in \mathbb{R}^m$, $m = u + l$ can be solved analytically if the constraint $f(x_i) = y_i$ is applied for all supervised samples. The labeled part of the vector f is therefore obvious. The unlabeled part of f is solved for by: